

Reversals and Sums

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, you have to process following operations:

1. reverse a subarray
2. calculate the sum of values in a subarray

Input

The first input line has two integers n and m : the size of the array and the number of operations. The array elements are numbered $1, 2, \dots, n$.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Finally, there are m lines that describe the operations. Each line has three integers t, a and b . If $t = 1$, you should reverse a subarray from a to b . If $t = 2$, you should calculate the sum of values from a to b .

Output

Print the answer to each operation where $t = 2$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 10^5$
- $0 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 3 2 1 3 4 5 3 4 4 2 2 4 1 3 6 2 2 4	8 9